Akshay Dadwal *Software Engineer*

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New Delhi, India

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Profile

A Software Engineer with 7 years of experience in Building solutions for complex B2B and B2C Systems.

While working with multiple startups, I have led and delivered several end-to-end products where my responsibilities were requirement gathering, creating system design docs, setup cloud infra, managing stakeholders, mentoring peers and code-reviews.

Skills

Django | Python | FastAPI | PostgreSQL | Celery | Redis | JavaScript | Node.js | Express.js | MongoDB | Kafka | Pub/Sub | AWS | GCP

Professional Experience

Detect Technologies Inc.

May 2025 – Present
Halifax,
Canada (Remote)

Software Development Engineer - III
Detect is an AI-powered utility asset management company which handles inspection of Transmission powerlines across Canada using guided drones to capture images and perform health-checks using in-house ***OpenCV models***.

Role and Responsibilities:

- Leading technical integration with ***Nova Scotia Power*** which is one of the largest electrical power companies in Canada.

Attentive.ai

Jan 2025 – Mar 2025
Noida, India

Software Engineer - III
Attentive.ai is an AI-driven SaaS platform which provides automated property measurements with in-house ***OpenCV models*** using aerial images, and ***Accelerate*** platform to manage day-to-day Operational activities for Landscaping companies in the US.

Role and Responsibilities:

- Collaborating with AI team to build a backend service which extract and parse text from PDF files and allows User(s) to perform search queries in real-time from Feathers(*Attentive's internal annotation platform*)

Jan 2022 – Dec 2024
Noida, India

Software Engineer - II
Role and Responsibilities:

- Leading the low-level and high-level design and development of backend service for *Operations module* using the Django, PostgreSQL, Celery and Redis, which facilitates User(s) to track revenue and cost of their Projects, Schedule tasks using *customized Calendar application and* a time-tracking application which computes employees' payroll

- Designed and developed an extensible User management service to authenticate and authorize public and private cross-service API calls to prevent data leakage among tenants by writting an abstraction over Django's permission framework to achieve Role and Attribute based Access-Control
- Integrated *New Relic* for Application monitoring, identified bottlenecks in APIs and DB queries, optimized Django ORM queries to reduce network I/Os and achieved ~20-40% improved latency in several APIs
- Contributed in *Residential microservice* using Express.js and MongoDB which integrates with in-house and 3rd-party *geospatial AI* services to provide automated measurements of residential properties
- Created low-level and high-level design for near real-time event processing application, and implemented within the *Residential microservice* using GCP Pub/Sub and Cloud functions, which executes several workflows i.e. notifications, background jobs and data aggregations
- Other responsibilities include, managing stakeholders, peer code-reviews, managing sprints and deployments and ensuring code-quality by enforcing standard Python and Django practices.

Renewbuy.com

Apr 2021 – Jan 2022
Gurgaon, India

Senior Software Engineer

Renewbuy.com is an Insurance aggregation platform which provides a streamlined and simplified Insurance buying experience for the End Customer and facilitates certified Insurance agents with their sales performance dashboard and other related services.

Role and Responsibilities:

- Designed and developed an OTP generation and verification microservice using FastAPI, which allows all internal services to delegate OTP related operations
- Developed and published a Python package for easier and maintainable cross-service Integration of OTP microservice
- Implemented monitoring and alerting on several backend services using DataDog and identified bottlenecks in the system i.e. slowest DB queries and APIs.
- Identified DB related issues and cross-service network calls, and created strategic document to optimize such APIs which resulted in ~30% improvement in overall latency
- Designed a near real-time data ingestion pipeline using Debezium, Kafka, AWS Redshift and Lambda to capture transactional data from multiple sources, transform into the desired form and load into OLAP backend in batches

Aug 2018 – Apr 2021
Gurgaon, India

Software Engineer

Role and responsibilities:

- Designed architecture revamp of Life Insurance platform to simplify integrations with 3rd-party Insurer services and Implemented new complex Product feature requirements which contributed significantly to company's revenue stream
- Designed and developed a Wallet/Payment Gateway aggregator microservice using Django to provide seamless payment experience and reduce dropout rates by ~15%
- Maintaining User onboarding and KYC platform, introducing new features and writing data extraction scripts for annual IRDAI Audits
- Conducted Security Audit for Self-managed and AWS Cloud Infra and Production Web Applications working with 3rd-party service vendor, identified severe security issues and patched those by collaborating with multiple teams
- Managing AWS Cloud Infra, pro-actively checking for bottlenecks in RDS instances, EC2 machines, AWS SQS and other cloud services

Awards

Nov 2024 — **Tech Trendsetter**
Attentive.ai
Got this Annual Award for exploring new technologies, improving system observability and creating technical documentation for peers.

Dec 2022 — **Innovation**
Attentive.ai
Received Annual Award for proactively identifying bottlenecks in the system, designing innovative solutions for complex business problems and implementing robust and scalable solutions.

Apr 2019 — **Rockstar Rookie**
D2C Consulting (*Renewbuy.com*)
Received Quarterly Award for performing exceptionally well as a new joiner, adapting quickly and making significant contributions on several Products.

Education

2014 – 2018 — **B.Tech - Information Technology**
Delhi, India — *HMRITM*

Personal Projects

dex-cli
A command-line Python tool to download and manage Manga/Comics from various platforms.
Features:
- Currently supports mangadex.org
- Uses multi-threading to avoid I/O blocks and perform concurrent file downloads
- Implemented Factory design pattern to easily integrate multiple manga provider platforms
- Unit test cases with ~60% code-coverage

Upcoming planned features:
- Integration with cloud storage services to improve accessibility
- Usage of optimal pagination strategy while consuming 3rd-party APIs to improve performance

Volunteer Experience

2023 – present — **Excercism**
Mentor
I have been contributing to Exercism community in my free time by helping out new Python programmers with their doubts/queries and constructive feedback in code-reviews.